EXP NO:2

Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

AIM:

Use sample data set sales-data.csv

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import csv

file_path='/sales_data.csv'
df = pd.read_csv(file_path)
print(df.head())
print(df.isnull().sum())
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())

df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
print(df.describe())

product_summary = df.groupby('Product').agg({
'Sales': 'sum',
'Quantity': 'sum'
}).reset_index()
print(product_summary)

plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
```

```
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('SalesOver Time')
plt.show()

pivot_table = df.pivot_table(values='Sales', index='Region',
columns='Product',
aggfunc=np.sum, fill_value=0)
print(pivot_table)
correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)

import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT:

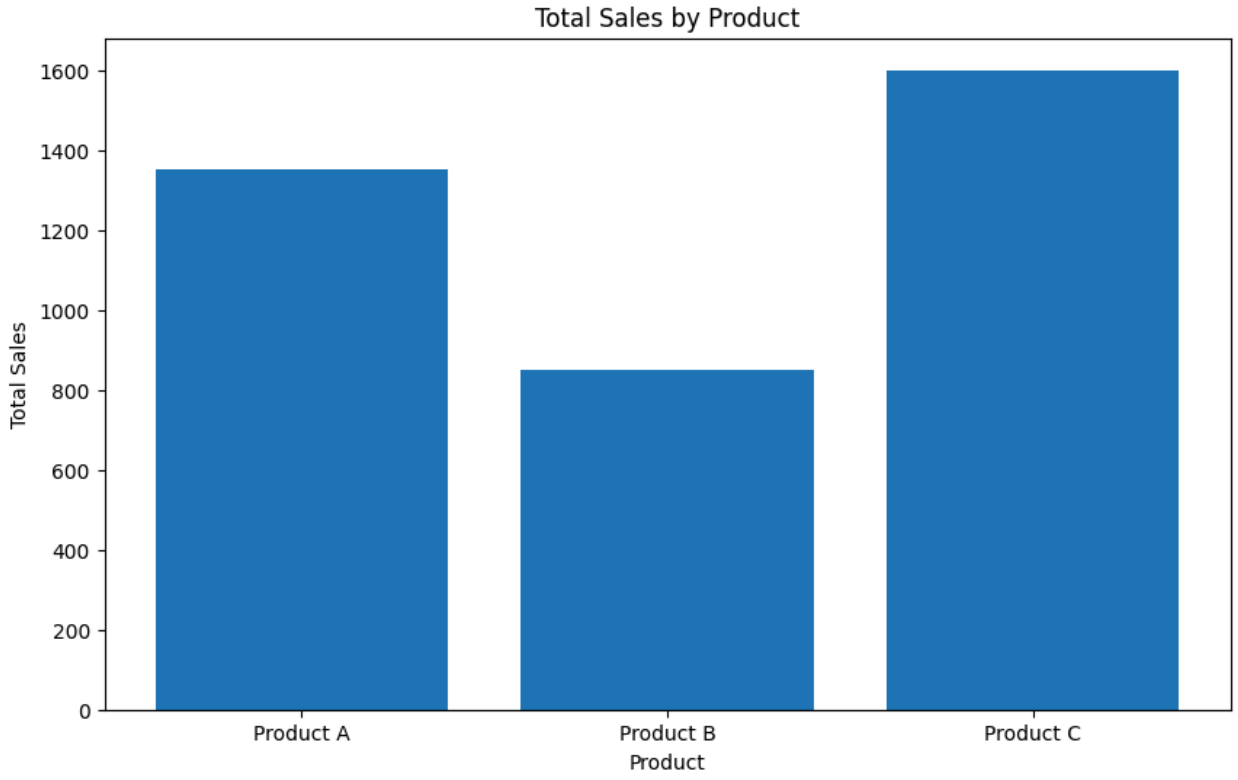
	Date	Product	Sales	Quantity	Region
0	01-01-2023	Product A	200	4	North
1	02-01-2023	Product B	150	3	South
2	03-01-2023	Product A	220	5	North
3	04-01-2023	Product C	300	6	East
4	05-01-2023	Product B	180	4	West

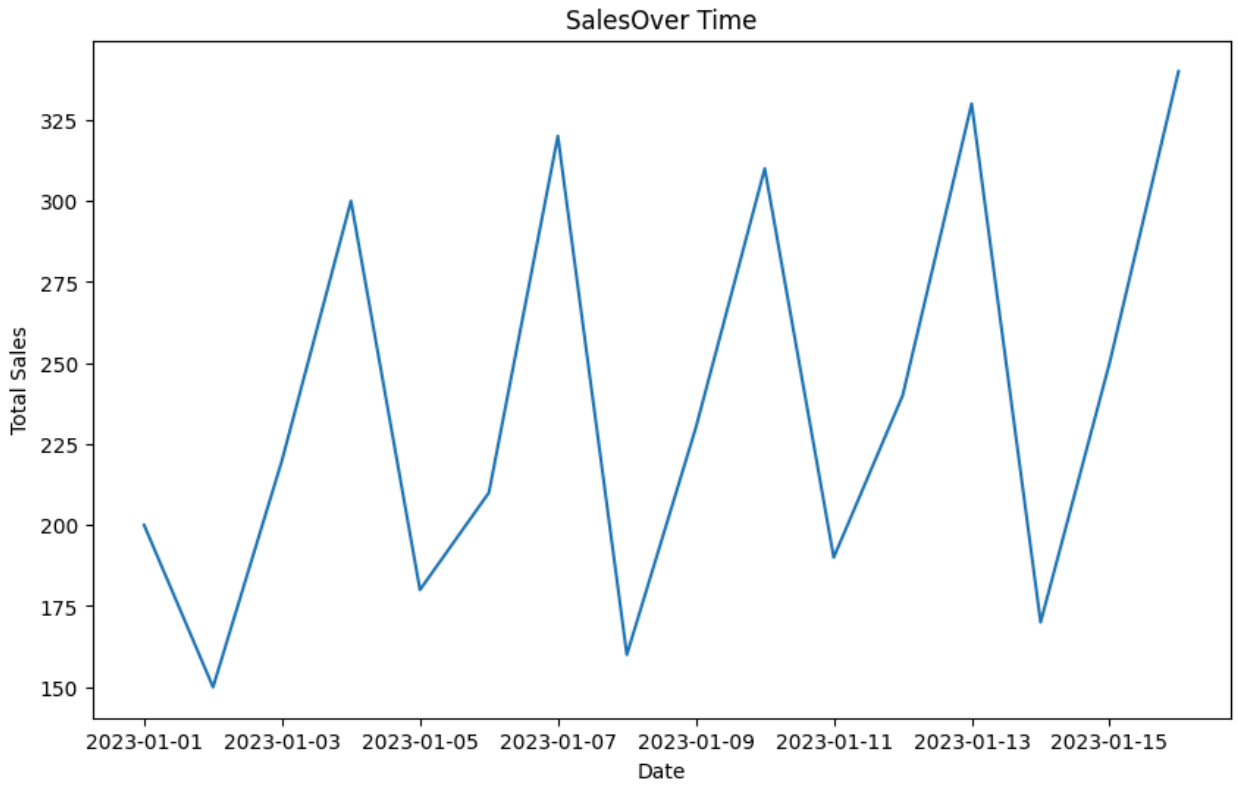
```
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
```

	Sales	Quantity
count	16.000000	16.000000
mean	237.500000	5.375000
std	64.031242	1.746425
min	150.000000	3.000000
25%	187.500000	4.000000
50%	225.000000	5.500000

75% 302.500000 7.000000
max 340.000000 8.000000

	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	17
2	Product C	1600	36

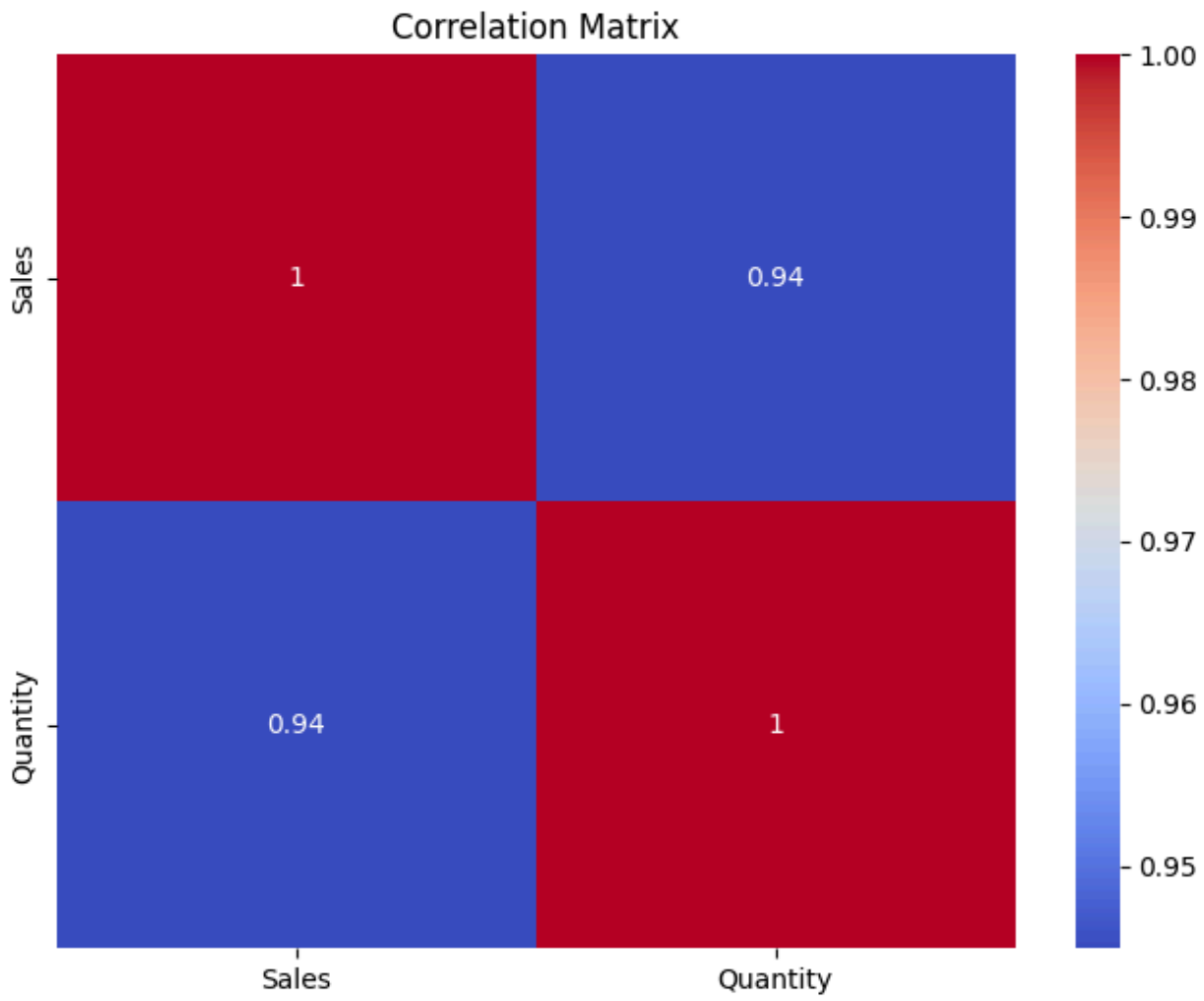




/tmp/ipykernel_1392/2537295789.py:37: FutureWarning: The provided callable <function sum at 0x7d378d57f380> is currently using DataFrameGroupBy.sum. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "sum" instead.
  pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',

Product	Product A	Product B	Product C
Region			
East	0	0	1600
North	1350	0	0
South	0	480	0
West	0	370	0

	Sales	Quantity
Sales	1.000000	0.944922
Quantity	0.944922	1.000000


Correlation Matrix
Sales
Quantity
1
0.94
0.94
1
Sales
Quantity
1.00
0.99
0.98
0.97
0.96
0.95